UTS UNIVERSITY OF TECHNOLOGY SYDNEY

Handbook

Subject

36120 - Advanced Machine Learning Application

8 Credit points

Faculty
TD School

Subject type
Subject

Study level
Postgraduate

Result type
Grade and marks

Total Workload hours
200

Subject description

This subject delves into advanced machine learning algorithms and machine learning operations (MLOps) concepts. You'll acquire skills to build, deploy, and manage models in real-world settings. The curriculum goes beyond basics, offering a deep dive into sophisticated algorithms and their practical uses through hands-on projects and case studies. A strong focus on deployment engineering prepares you to navigate the complexities of integrating machine learning models into production.

Availabilities

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Learning outcomes

Upon completion of this subject, graduates will be able to:

1.	Implement and evaluate advanced machine learning techniques to real-world problems
2.	Design and implement robust and efficient systems for model deployment
3.	Explain knowledge, skills and best practices required from different disciplines for managing the lifecycle of machine learning models
4.	Identify business outcomes (benefits, risks, cost) and legal and ethical implications (fairness, accountability, privacy and transparency) of deploying machine learning models
5.	Apply effective collaboration and communication skills necessary for working in interdisciplinary teams

Learning and teaching activities

Blend of online and face to face activities: The subject is offered through a series of teaching sessions which blend online and face-to-face learning. Students learn through interactive lectures and classroom activities making use of the subject materials on canvas. They also engage in individual and collaborative learning activities to understand and apply text analysis techniques in diverse settings.
Authentic problem based learning: This subject offers a range of authentic data science problems to solve that will help develop students' text analysis skills. They work on real world data analysis problems for broad areas of interest using unstructured data and contemporary techniques.
Collaborative work: Group activities will enable students to leverage peer-learning and demonstrate effective team participation, as well as learning to work in professional teams with an appreciation of diverse perspectives on data science and innovation.
Future-oriented strategies: Students will be exposed to contemporary learning models using speculative thinking, ethical and human-centered approaches as well as reflection. Electronic portfolios will be used to curate, consolidate and provide evidence of learning and development of course outcomes, graduate attributes and professional evolution. Formative feedback will be offered with all assessment activities for successful engagement.

Assessment information

Click on Select availability in the top right of screen and choose from the location, attendance mode and session options.

Requisites

Requisite detail

Before you enrol in this subject, you must ensure you have met its requisites. The requisites for this subject can be viewed on its access conditions page.

To learn more about requisites, please visit the How to use the Handbook guide.

Apply now **Important dates** **Fee calculator**

UTS acknowledges the Gadigal people of the Eora Nation, the Boorooberongal people of the Dharug Nation, the Bidiagal people and the Gamaygal people, upon whose ancestral lands our university stands. We would also like to pay respect to the Elders both past and present, acknowledging them as the traditional custodians of knowledge for these lands.

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